Lecture 16 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that prints the contents of a text file to the screen, numbering each line.

For example, suppose groceries.txt contains the following:

```
eggs
milk
coffee
more coffee
chocolate
```

Then:

```
What file would you like to print? groceries.txt

1. eggs
2. milk
3. coffee
4. more coffee
5. chocolate
```

Sample Solution

```
filename = input('What file would you like to print? ')
print()

input_file = open(filename, 'r')  # 'r' means open the file for reading only

line_number = 1
for line in input_file:
    print(f'{line_number}. {line.strip()}')  # strip() removes the newline characte
    r
    line_number += 1

input_file.close()
```

Question 2 (q2_sol.py)

Write a program that finds and prints all the English words that don't have a vowel. The vowels are a, e, i, o, u, and y, and use words.txt as the list of English words.

```
1. crwth
2. crwths
3. cwm
4. cwms
5. nth
...
14. tsktsks
```

Sample Solution

```
def has_vowel(word):
    for vowel in 'aeiouy':
        if vowel in word:
            return True
    return False

#
# open the words file and read it line by line
#
words_file = open('words.txt', 'r')
num = 0
for word in words_file:
    word = word.strip()
    if not has_vowel(word):
        num += 1
        print(f'{num}. {word}')

words_file.close()
```

Question 3 (q3_sol.py)

Write a program that reads a string the user and then prints the letter and index value in the style shown.

```
Enter a string: hello

s = 'hello'
s[0] is 'h'
s[1] is 'e'
s[2] is 'l'
s[3] is 'l'
s[4] is 'o'
```

Sample Solution

```
s = input('Enter a string: ')
print()
print(f"s = '{s}'")
for i in range(len(s)):
    letter = s[i]
    print(f"s[{i}] is '{letter}'")
```

Question 4 (q4_sol.py)

Write a program that repeatedly asks the user for a date in the format YYYY-MM-DD and prints the same date in the format DD/MM/YYYY. The program should end with the user pressing <enter> with no date entered.

You can assume the user always types the date in the correct format.

```
Enter a date (YYYY-MM-DD): 2026-10-25
25/10/2026
Enter a date (YYYY-MM-DD): 1992-08-14
14/08/1992
Enter a date (YYYY-MM-DD): <enter>
Done!
```

Sample Solution

```
while True:
    date = input('Enter a date (YYYY-MM-DD): ')
    if date == '':
        break
    year = date[0:4]
    month = date[5:7]
    day = date[8:10]
    print(f'{day}/{month}/{year}')
```

Question 5 (q5_sol.py)

Write a program that repeatedly asks the user to enter a word, and then prints the plural of the word (as described below). The program should stop when the user enters an empty string.

Write a function called pluralize(s) that takes a string s and returns a plural version of the string according to these rules:

- If the string ends with 's', do nothing.
- If the string ends with 'y', replace it with 'ies'.
- Otherwise, add an 's' to the end of the string.

Sample Run

```
Enter a word: cat
cats
Enter a word: baby
babies
Enter a word: penguins
penguins
Enter a word: <enter>
Done!
```

Sample Solution

```
def pluralize(s):
    if s[-1] == 's':
        return s
    elif s[-1] == 'y':
        return s[:-1] + 'ies' # remove the last letter and add 'ies'
    else:
        return s + 's' # add 's' to the end of the string

while True:
    word = input('Enter a word: ')
    if word == '':
        break
    print(pluralize(word))
print('Done!')
```